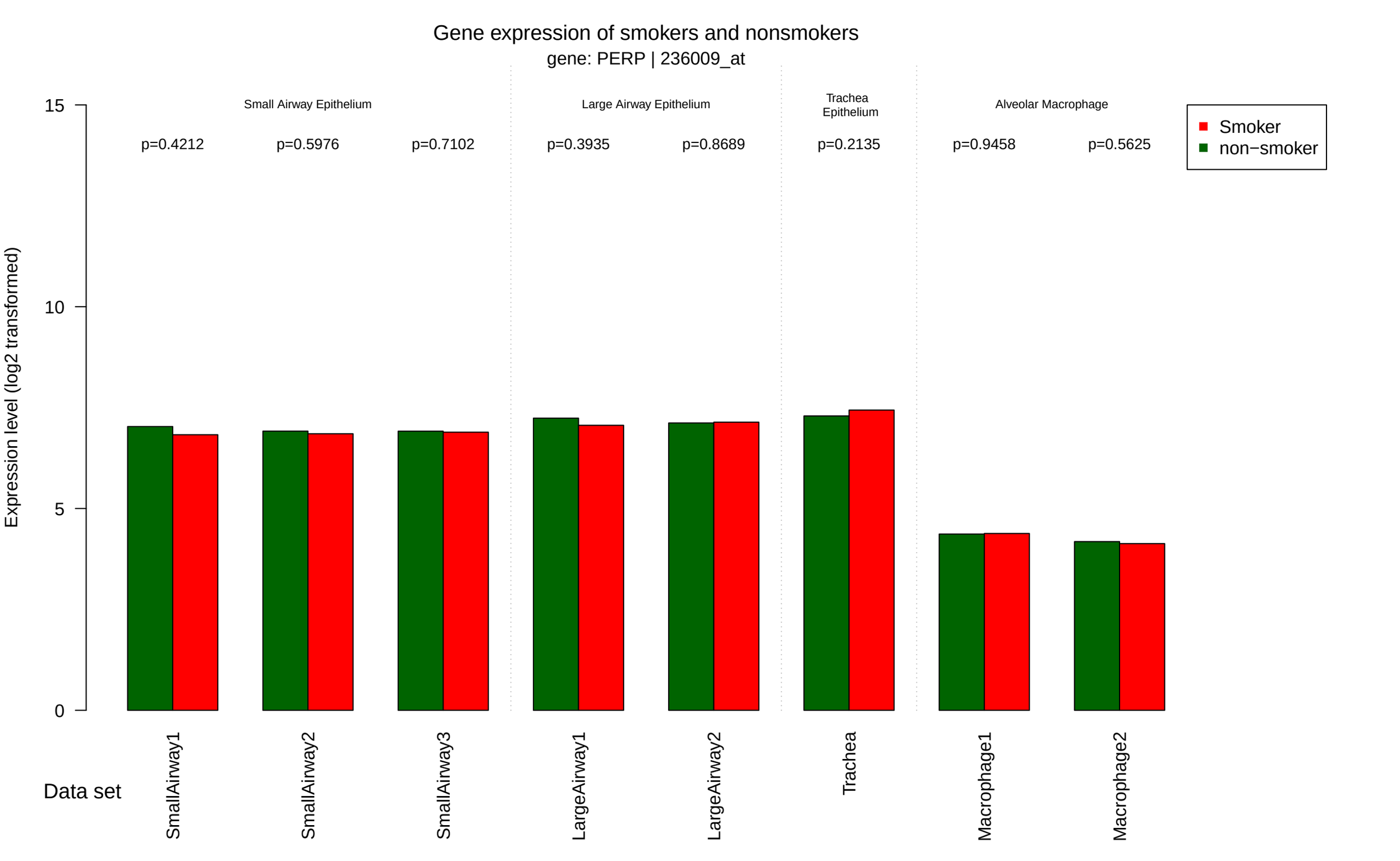
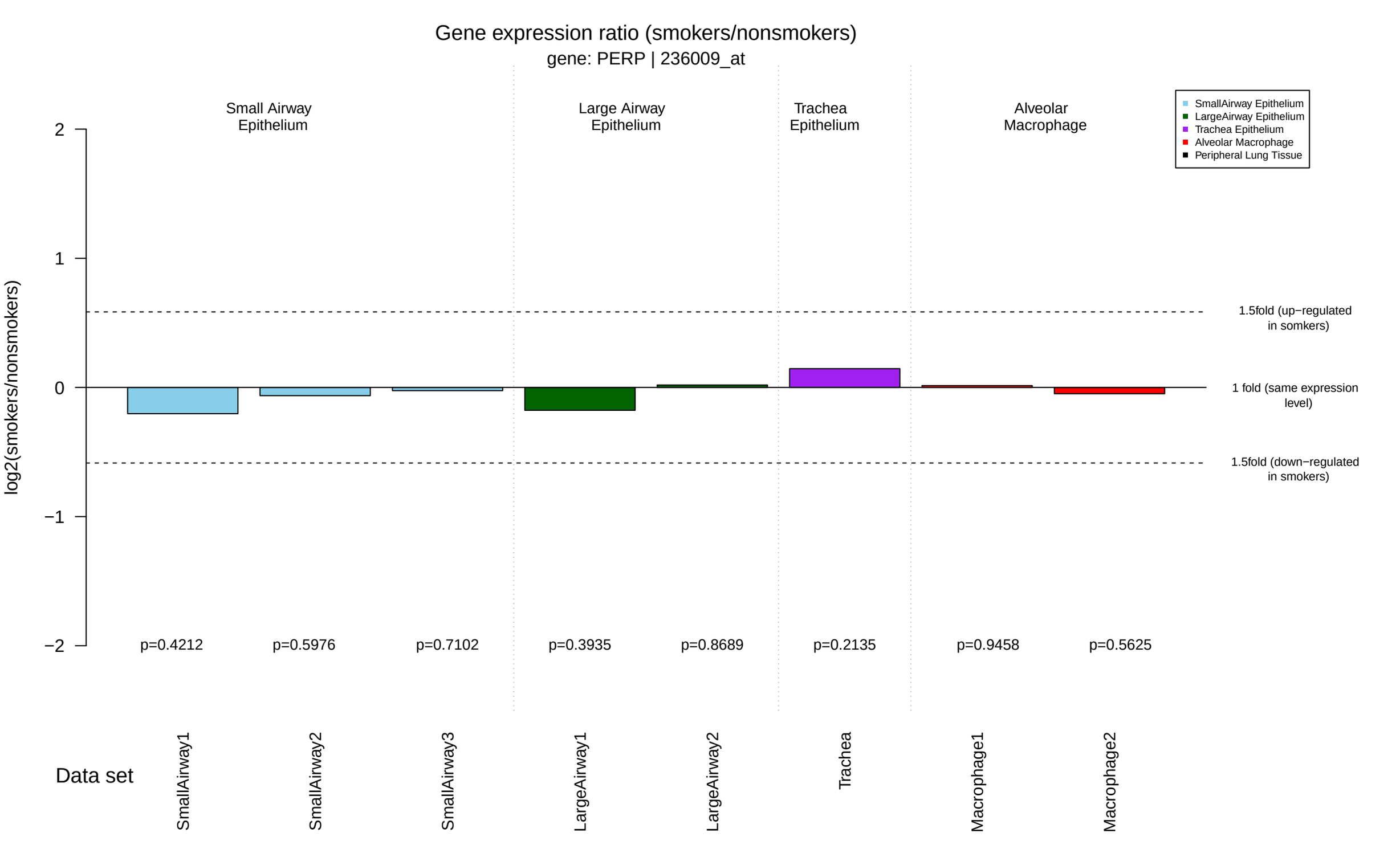
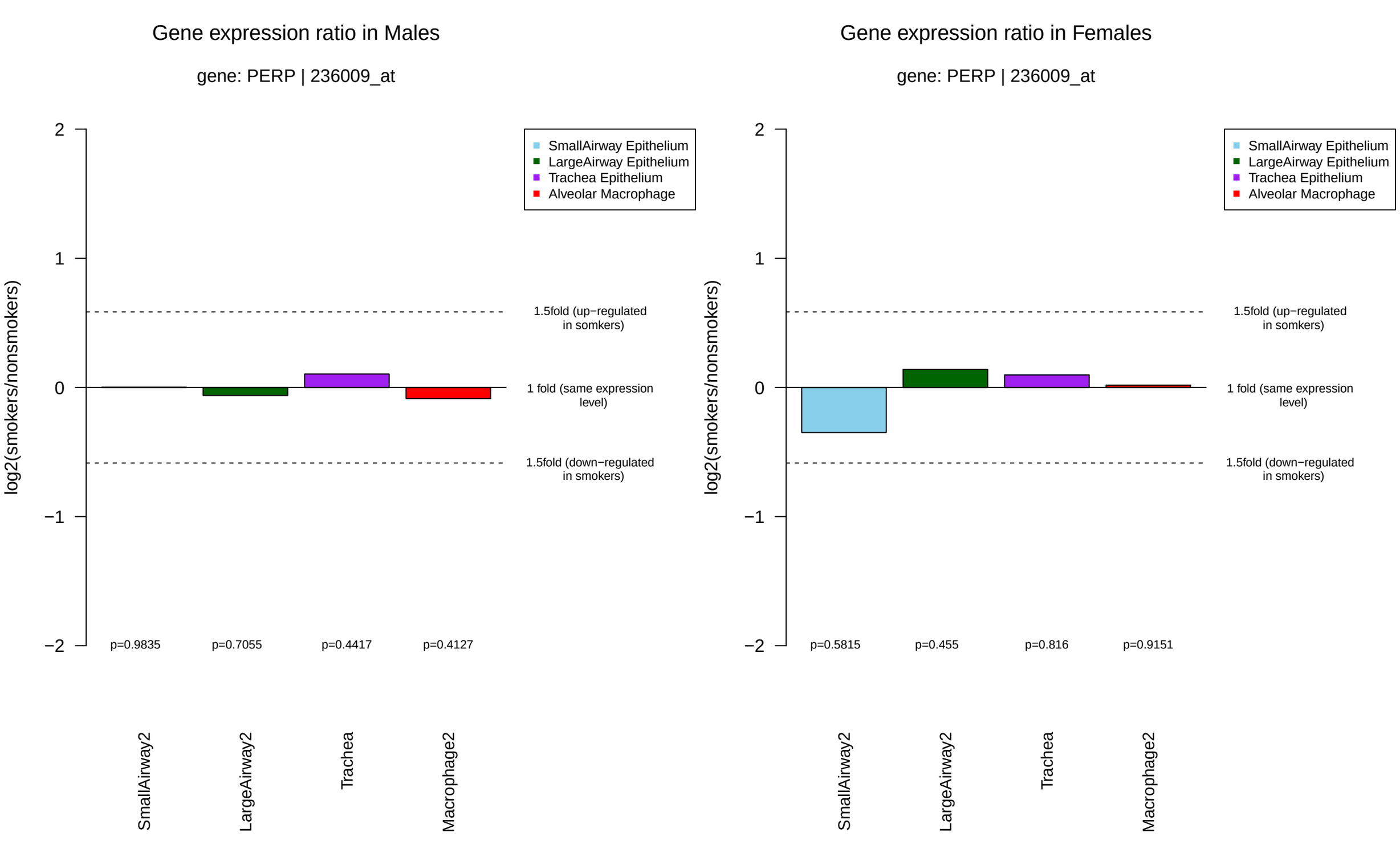
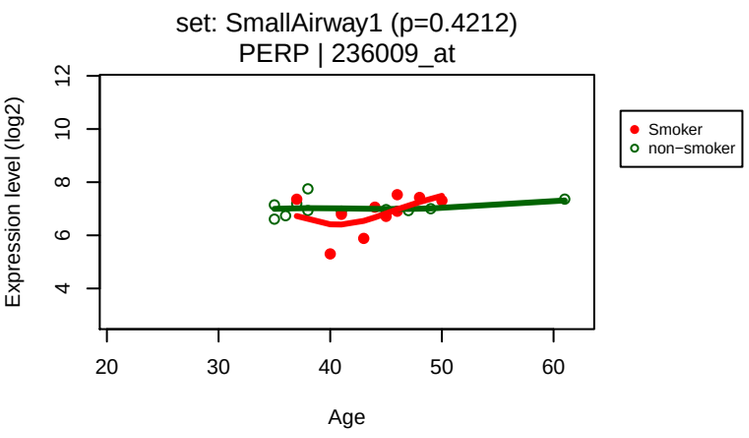
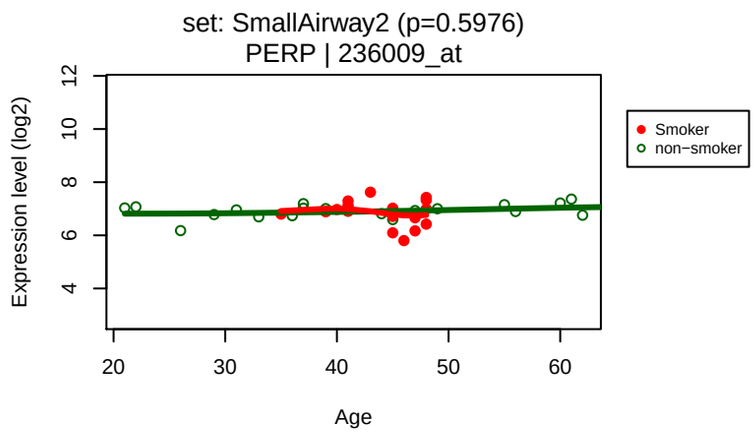
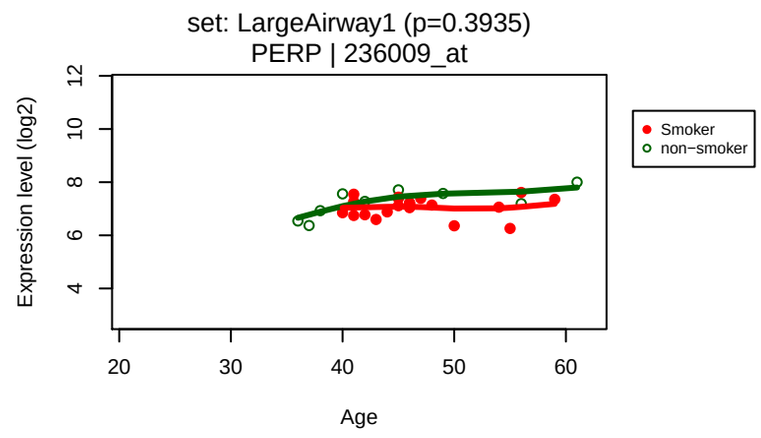

Gene expression of smokers and nonsmokers
gene: PERP | 236009_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4212
p=0.5976
p=0.7102
p=0.3935
p=0.8689
p=0.2135
p=0.9458
p=0.5625
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

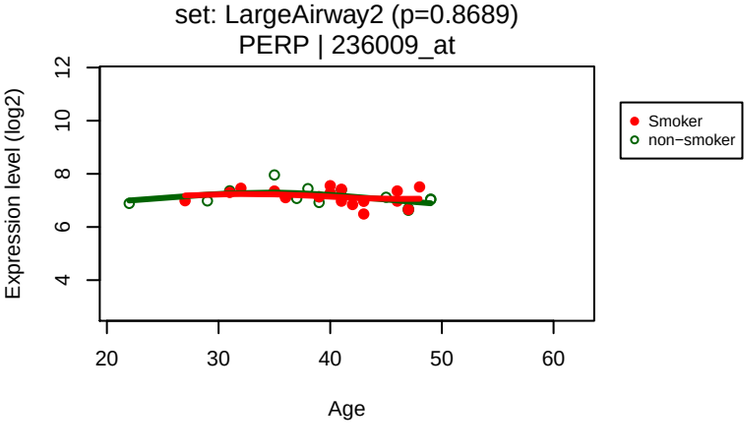

Gene expression ratio (smokers/nonsmokers)
gene: PERP | 236009_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4212
p=0.5976
p=0.7102
p=0.3935
p=0.8689
p=0.2135
p=0.9458
p=0.5625
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

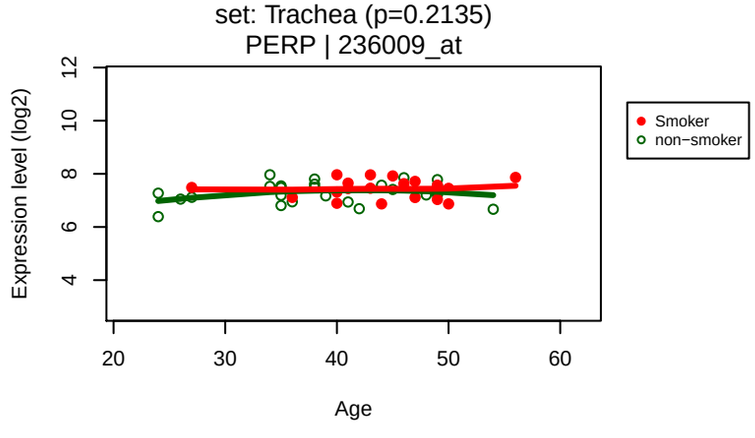

Gene expression ratio in Males
gene: PERP | 236009_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9835
p=0.7055
p=0.4417
p=0.4127
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PERP | 236009_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5815
p=0.455
p=0.816
p=0.9151
SmallAirway2
LargeAirway2
Trachea
Macrophage2

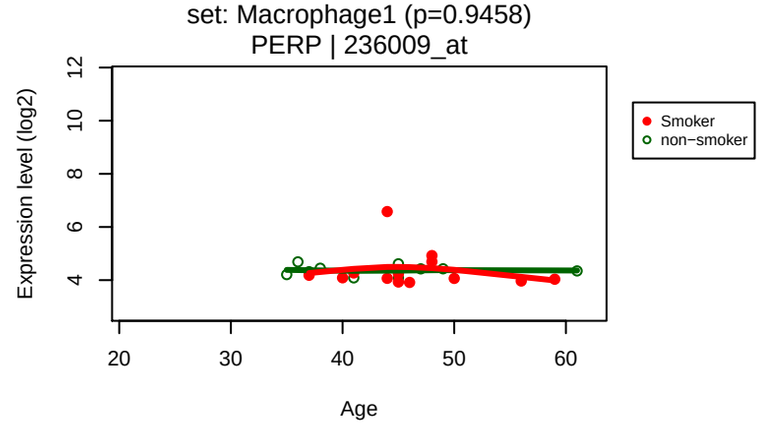

set: SmallAirway1 (p=0.4212)
PERP | 236009_at
Expression level (log2)
Age
Smoker
non−smoker

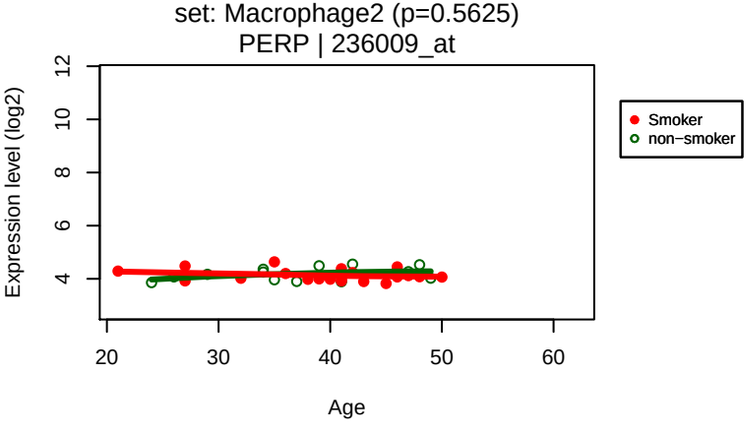

set: SmallAirway2 (p=0.5976)
PERP | 236009_at
Expression level (log2)
Age
Smoker
non−smoker

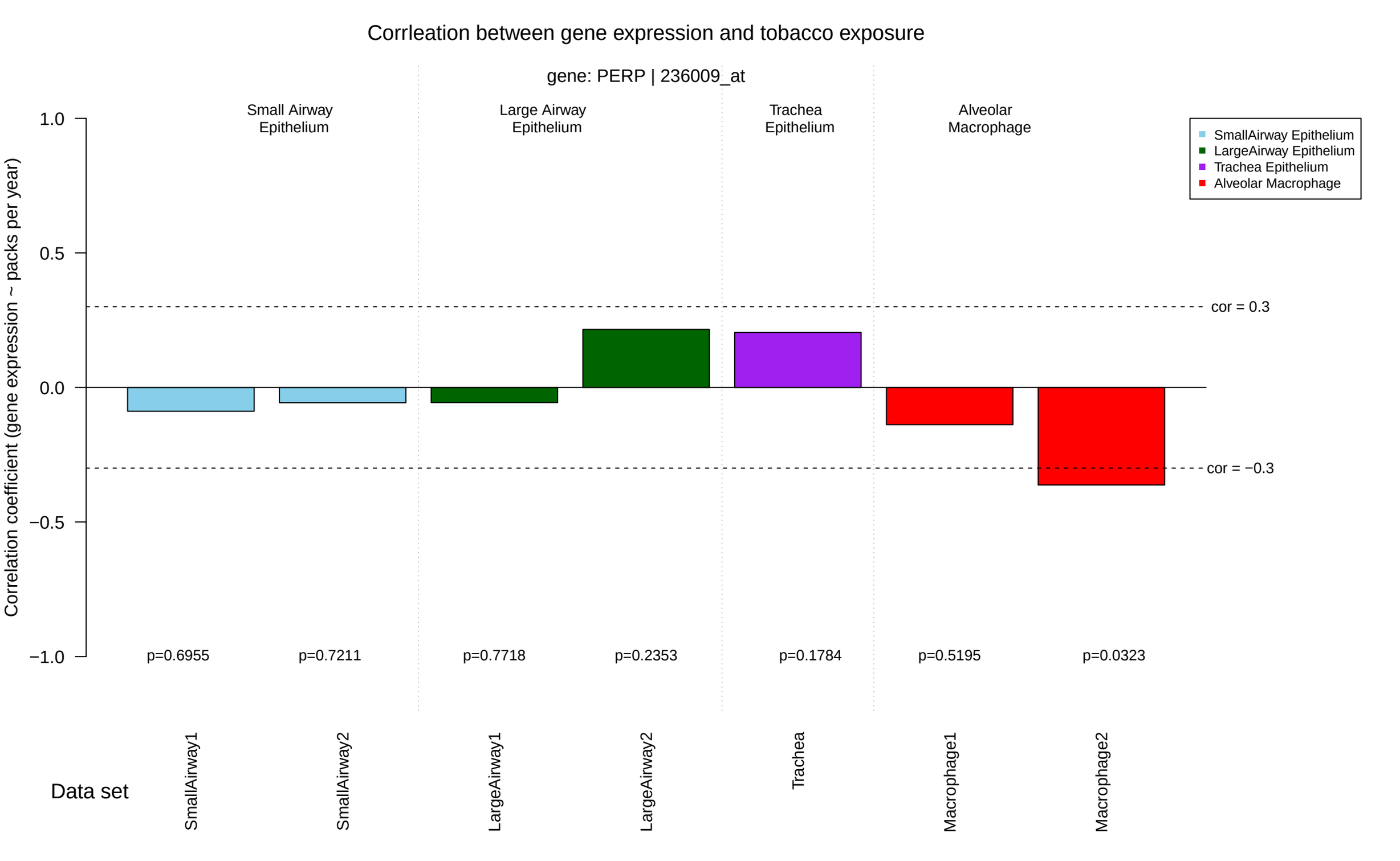

set: LargeAirway1 (p=0.3935)
PERP | 236009_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8689)
PERP | 236009_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2135)
PERP | 236009_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9458)
PERP | 236009_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5625)
PERP | 236009_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PERP | 236009_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6955
p=0.7211
p=0.7718
p=0.2353
p=0.1784
p=0.5195
p=0.0323
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2